

INFORMATION TECHNOLOGY

Complete Coursebook

80-page academic course for senior secondary / Terminale level

Core areas

Hardware - Software - Office Applications - Databases - Networks - Cybersecurity - Web - IT Support - Cloud - Emerging Technologies

Language: English explanations

Format: Lessons, tables, examples, practical tasks, revision and answer guidance

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Lesson 1: What Is Information Technology?

Foundations of Information Technology

Information Technology (IT) is the use of computer systems, software, networks and digital services to create, process, store, secure and exchange information. IT supports education, banking, health care, government, science, transport and business. Unlike Computer Science, which often focuses on algorithms and computation, IT emphasizes selecting, deploying, maintaining and using technology in real organizations.

Learning objectives

- Understand the main purpose of what is information technology?.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

hardware	software	data	users	procedures
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **What Is Information Technology?** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Explain the difference between IT and Computer Science using one real-life example.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 2: Computer Systems and Components

Foundations of Information Technology

A computer system combines hardware, software, data and people. Hardware includes the processor, memory, storage, input devices and output devices. Software tells the hardware what to do. Data becomes useful information after processing. Users and procedures ensure that technology is used correctly and consistently.

Learning objectives

- Understand the main purpose of computer systems and components.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

CPU	RAM	SSD	input	output
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Computer Systems and Components** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Identify five components in a school computer laboratory and state the role of each.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 3: Data, Information and Knowledge

Foundations of Information Technology

Data are raw facts such as numbers, names or measurements. Information is data that has been organized and interpreted. Knowledge is the understanding used to make decisions. Good IT systems transform accurate data into useful information while preserving meaning, context and quality.

Learning objectives

- Understand the main purpose of data, information and knowledge.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

accuracy	relevance	timeliness	completeness	consistency
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Data, Information and Knowledge** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Give an example of raw data becoming information and then knowledge.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 4: Digital Devices and Platforms

Foundations of Information Technology

Digital technology includes desktop computers, laptops, tablets, smartphones, servers, embedded devices and Internet of Things devices. Each platform differs in performance, mobility, energy use, cost and purpose. Choosing a device requires matching technical capability to user needs.

Learning objectives

- Understand the main purpose of digital devices and platforms.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

desktop	laptop	tablet	smartphone	server
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Digital Devices and Platforms** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Choose the best device for a field worker and justify your choice.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 5: Health, Safety and Ergonomics

Foundations of Information Technology

Safe IT use includes correct posture, suitable lighting, regular breaks, secure cables and protection from electrical hazards. Ergonomics is the design of workspaces and tools to improve comfort, efficiency and safety. Long periods of poor posture can contribute to eye strain and musculoskeletal discomfort.

Learning objectives

- Understand the main purpose of health, safety and ergonomics.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

ergonomics	posture	lighting	breaks	electrical safety
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Health, Safety and Ergonomics** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Design a safe workstation checklist for a computer user.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 6: The CPU and Processing

Hardware and Computer Architecture

The central processing unit executes instructions and controls system operations. It includes the control unit, arithmetic logic unit and registers. Clock speed, number of cores, cache and processor architecture influence performance, although real performance also depends on software, memory and workload.

Learning objectives

- Understand the main purpose of the cpu and processing.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

ALU	control unit	register	cache	core
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **The CPU and Processing** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Why can a processor with a lower clock speed sometimes outperform another processor?
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 7: Memory: RAM, ROM and Cache

Hardware and Computer Architecture

RAM stores programs and data currently in use and is volatile, meaning its contents are lost when power is removed. ROM stores permanent instructions. Cache is small, fast memory close to the CPU. A balanced system needs enough RAM and efficient memory access for its applications.

Learning objectives

- Understand the main purpose of memory: ram, rom and cache.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

RAM	ROM	cache	volatile	capacity
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Memory: RAM, ROM and Cache** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Compare RAM, ROM and cache in a three-column table.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 8: Storage Technologies

Hardware and Computer Architecture

Storage keeps data for long-term use. Solid-state drives use flash memory and have no moving mechanical parts. Hard disk drives use magnetic platters and usually provide larger capacity at lower cost. Optical media, USB flash drives, memory cards and cloud storage provide additional options.

Learning objectives

- Understand the main purpose of storage technologies.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

SSD	HDD	flash	optical	cloud
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Storage Technologies** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Recommend storage for a video editor, a student and a long-term archive.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 9: Input Devices

Hardware and Computer Architecture

Input devices allow users or sensors to enter data. Examples include keyboards, mice, touchscreens, microphones, cameras, scanners, biometric readers and sensors. The best input method depends on speed, accuracy, environment, accessibility and the type of data being captured.

Learning objectives

- Understand the main purpose of input devices.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

keyboard	scanner	camera	sensor	biometric
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Input Devices** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Give two advantages and two limitations of biometric input.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 10: Output Devices

Hardware and Computer Architecture

Output devices present processed information. Monitors, printers, speakers, projectors and actuators are common examples. Important characteristics include resolution, color accuracy, speed, durability, energy consumption and compatibility with the intended task.

Learning objectives

- Understand the main purpose of output devices.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

monitor	printer	speaker	projector	actuator
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Output Devices** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Which output device is most suitable for a large classroom presentation, and why?
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 11: Ports, Interfaces and Peripherals

Hardware and Computer Architecture

Computer systems connect to peripherals through interfaces such as USB, HDMI, Ethernet, audio ports, Bluetooth and Wi-Fi. Standards help devices communicate reliably. Users should consider data speed, power requirements, physical connectors and compatibility.

Learning objectives

- Understand the main purpose of ports, interfaces and peripherals.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

USB	HDMI	Ethernet	Bluetooth	Wi-Fi
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Ports, Interfaces and Peripherals** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Match five peripherals with the most appropriate connection method.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 12: System Software and Application Software

Software and Operating Systems

System software manages hardware and provides services for other programs. Operating systems, device drivers and utilities are examples. Application software helps users perform tasks such as writing documents, analyzing data, editing images or managing customers.

Learning objectives

- Understand the main purpose of system software and application software.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

system software	application software	driver	utility	platform
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **System Software and Application Software** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Classify ten programs as system or application software.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 13: Operating System Functions

Software and Operating Systems

An operating system manages processes, memory, files, devices, security and the user interface. It also provides services that applications use. Popular desktop, mobile and server operating systems differ in licensing, hardware support, ecosystem and administration tools.

Learning objectives

- Understand the main purpose of operating system functions.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

process	memory	file system	device management	security
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Operating System Functions** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Explain five responsibilities of an operating system.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 14: User Interfaces

Software and Operating Systems

Graphical user interfaces use windows, icons, menus and pointers, while command-line interfaces rely on typed commands. Touch and voice interfaces are increasingly common. The best interface depends on task, expertise, accessibility and environment.

Learning objectives

- Understand the main purpose of user interfaces.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

GUI	CLI	touch	voice	accessibility
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **User Interfaces** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Compare GUI and CLI for a beginner and for a system administrator.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 15: File Management

Software and Operating Systems

Files should be named clearly, organized in logical folders and backed up. File extensions indicate formats such as .docx, .pdf, .jpg or .csv. Good file management reduces duplication, improves retrieval and supports collaboration.

Learning objectives

- Understand the main purpose of file management.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

folder	filename	extension	path	backup
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **File Management** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Create a folder structure for a school administration office.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 16: Software Installation and Updates

Software and Operating Systems

Software installation may require checking system requirements, licensing, storage space and permissions. Updates can add features, fix bugs and patch security vulnerabilities. Organizations often test updates before wide deployment to reduce disruption.

Learning objectives

- Understand the main purpose of software installation and updates.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

installation	update	patch	license	compatibility
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Software Installation and Updates** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Why should important systems test updates before installing them everywhere?
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 17: Open Source, Proprietary and Cloud Software

Software and Operating Systems

Open-source software provides source code under licenses that allow varying degrees of use and modification. Proprietary software is controlled by its owner. Cloud software is delivered over networks, often through a subscription. Each model has different costs, control, support and security implications.

Learning objectives

- Understand the main purpose of open source, proprietary and cloud software.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

open source	proprietary	cloud	license	subscription
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Open Source, Proprietary and Cloud Software** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Compare open-source and proprietary software for a small organization.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 18: Word Processing

Productivity and Office Applications

Word processors help create, edit and format text documents. Features include styles, tables, headers, footers, references, mail merge and collaboration. Professional documents use consistent formatting, clear structure and appropriate file formats.

Learning objectives

- Understand the main purpose of word processing.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

styles	tables	header	mail merge	collaboration
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Word Processing** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Create a checklist for producing a professional two-page report.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 19: Spreadsheets

Productivity and Office Applications

Spreadsheets organize data in rows and columns and can perform calculations using formulas and functions. They support sorting, filtering, charts and data analysis. Relative and absolute cell references affect how formulas behave when copied.

Learning objectives

- Understand the main purpose of spreadsheets.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

formula	function	relative reference	absolute reference	chart
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Spreadsheets** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Write examples using SUM, AVERAGE, IF and an absolute reference.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 20: Presentations

Productivity and Office Applications

Presentation software combines text, images, diagrams, charts, audio and video. Effective slides use concise wording, strong hierarchy and visual consistency. A presentation should support the speaker rather than reproduce a full report.

Learning objectives

- Understand the main purpose of presentations.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

slide	layout	theme	chart	speaker notes
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Presentations** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. List five rules for clear presentation design.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 21: Desktop Publishing and Graphics

Productivity and Office Applications

Desktop publishing software is used to design brochures, posters, newsletters and other page-based documents. Raster graphics are made of pixels, while vector graphics are based on paths and shapes. The choice affects scaling, editing and print quality.

Learning objectives

- Understand the main purpose of desktop publishing and graphics.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

DTP	raster	vector	resolution	layout
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Desktop Publishing and Graphics** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Explain when vector graphics are preferable to raster graphics.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 22: Collaboration Tools

Productivity and Office Applications

Digital collaboration tools support shared documents, messaging, meetings, task management and version control. Good collaboration requires clear permissions, file naming, communication rules and attention to privacy.

Learning objectives

- Understand the main purpose of collaboration tools.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

shared document	version	permission	meeting	task
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Collaboration Tools** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Propose three rules for a team working on a shared online document.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 23: Introduction to Databases

Data Management and Databases

A database stores related data in a structured way. Database management systems help users create, query, update and secure data. Databases reduce duplication and support reliable access when designed carefully.

Learning objectives

- Understand the main purpose of introduction to databases.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

database	DBMS	table	record	field
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Introduction to Databases** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Define table, record and field using a student database example.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 24: Relational Databases

Data Management and Databases

Relational databases organize data into tables connected by keys. A primary key uniquely identifies a record, while a foreign key links tables. Relationships can be one-to-one, one-to-many or many-to-many.

Learning objectives

- Understand the main purpose of relational databases.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

primary key	foreign key	relationship	table	record
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Relational Databases** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Design two related tables for students and classes.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 25: Data Types and Validation

Data Management and Databases

Fields should use suitable data types such as text, integer, decimal, date or Boolean. Validation rules prevent invalid entries and improve data quality. Examples include required fields, range checks and format checks.

Learning objectives

- Understand the main purpose of data types and validation.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

data type	validation	range check	required	format
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Data Types and Validation** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Create validation rules for age, email and exam score.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 26: SQL Fundamentals

Data Management and Databases

Structured Query Language is used to work with relational databases. SELECT retrieves data, INSERT adds records, UPDATE changes values and DELETE removes records. WHERE filters rows and ORDER BY controls sorting.

Learning objectives

- Understand the main purpose of sql fundamentals.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

SELECT	WHERE	ORDER BY	INSERT	UPDATE
--------	-------	----------	--------	--------

Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **SQL Fundamentals** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Write a query to list students with scores above 70, highest score first.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 27: Database Design and Normalization

Data Management and Databases

Normalization organizes tables to reduce unnecessary duplication and update problems. Good design separates independent entities, identifies keys and creates appropriate relationships. The goal is consistent data that can be maintained efficiently.

Learning objectives

- Understand the main purpose of database design and normalization.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

normalization	entity	redundancy	key	relationship
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Database Design and Normalization** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Explain why storing a teacher phone number repeatedly in every student record can be problematic.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 28: Data Quality and Governance

Data Management and Databases

Data governance defines responsibilities and rules for collecting, storing, using, sharing and deleting data. Quality dimensions include accuracy, completeness, consistency, validity and timeliness. Governance also addresses privacy and accountability.

Learning objectives

- Understand the main purpose of data quality and governance.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

governance	quality	privacy	retention	accountability
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Data Quality and Governance** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Draft four rules for handling student records responsibly.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 29: Network Fundamentals

Networks and the Internet

A computer network connects devices so they can exchange data and share resources. Networks may be local, metropolitan or wide-area. Network design considers speed, reliability, security, cost and scale.

Learning objectives

- Understand the main purpose of network fundamentals.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

LAN	WAN	node	bandwidth	latency
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Network Fundamentals** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Compare a LAN and a WAN using a school and the Internet as examples.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 30: Network Hardware

Networks and the Internet

Switches connect devices within a local network, routers forward traffic between networks and wireless access points connect Wi-Fi devices. Modems connect local equipment to certain communication services. Firewalls filter traffic according to rules.

Learning objectives

- Understand the main purpose of network hardware.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

switch	router	access point	modem	firewall
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Network Hardware** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. State the role of a switch, router and access point.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 31: IP Addresses and Protocols

Networks and the Internet

Internet Protocol addresses identify devices or interfaces on IP networks. Protocols define rules for communication. TCP supports reliable data transfer, HTTP and HTTPS support web communication, DNS translates names, and DHCP can assign network settings automatically.

Learning objectives

- Understand the main purpose of ip addresses and protocols.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

IP	TCP	HTTP	DNS	DHCP
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **IP Addresses and Protocols** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Explain what happens when a user enters a website name into a browser.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 32: Wireless Networking

Networks and the Internet

Wireless networks use radio signals rather than cables. Wi-Fi offers mobility but may be affected by distance, interference and obstacles. Security measures include strong encryption, secure passwords and careful network configuration.

Learning objectives

- Understand the main purpose of wireless networking.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

Wi-Fi	SSID	encryption	interference	coverage
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Wireless Networking** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Suggest four ways to improve a weak school Wi-Fi network.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 33: Internet Services

Networks and the Internet

The Internet supports web browsing, email, cloud applications, voice and video communication, streaming and many other services. Each service depends on protocols, servers, clients and network infrastructure.

Learning objectives

- Understand the main purpose of internet services.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

web	email	cloud	streaming	VoIP
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Internet Services** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Describe the difference between the Internet and the World Wide Web.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 34: Cloud Computing

Networks and the Internet

Cloud computing provides computing resources such as storage, servers and software over networks. Common service models include infrastructure, platform and software services. Benefits include scalability and remote access, while risks include dependency, privacy and service availability.

Learning objectives

- Understand the main purpose of cloud computing.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

IaaS	PaaS	SaaS	scalability	availability
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Cloud Computing** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Give one example use case for IaaS, PaaS and SaaS.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 35: Cybersecurity Fundamentals

Cybersecurity and Digital Safety

Cybersecurity protects systems, networks and data from unauthorized access, damage or disruption. The confidentiality, integrity and availability model is often used to describe core security goals.

Learning objectives

- Understand the main purpose of cybersecurity fundamentals.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

confidentiality	integrity	availability	risk	control
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Cybersecurity Fundamentals** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Give one security control for confidentiality, integrity and availability.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 36: Passwords and Authentication

Cybersecurity and Digital Safety

Strong authentication reduces unauthorized access. Long unique passwords, password managers and multi-factor authentication are stronger than reused short passwords. Biometrics can provide convenience but should be implemented with appropriate privacy safeguards.

Learning objectives

- Understand the main purpose of passwords and authentication.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

password	MFA	biometric	credential	authentication
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Passwords and Authentication** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Explain why multi-factor authentication improves security.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 37: Malware and Social Engineering

Cybersecurity and Digital Safety

Malware includes viruses, worms, trojans, ransomware and spyware. Social engineering manipulates people into revealing information or performing unsafe actions. Technical controls are important, but user awareness is also essential.

Learning objectives

- Understand the main purpose of malware and social engineering.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

virus	ransomware	spyware	phishing	social engineering
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Malware and Social Engineering** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. List five signs that an email may be a phishing attempt.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 38: Backups and Recovery

Cybersecurity and Digital Safety

Backups protect against accidental deletion, hardware failure, ransomware and disasters. The 3-2-1 approach keeps multiple copies on different media, including one copy stored separately. Recovery procedures should be tested.

Learning objectives

- Understand the main purpose of backups and recovery.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

backup	restore	3-2-1	recovery	archive
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Backups and Recovery** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Design a weekly backup plan for a small office.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 39: Privacy and Data Protection

Cybersecurity and Digital Safety

Privacy concerns how personal information is collected, used and shared. Organizations should collect only necessary data, limit access, protect stored information and define retention periods. Users should understand permissions and sharing settings.

Learning objectives

- Understand the main purpose of privacy and data protection.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

privacy	personal data	consent	access control	retention
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Privacy and Data Protection** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Give four practical ways a school can protect student privacy.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 40: Incident Response and Security Policy

Cybersecurity and Digital Safety

An incident response process prepares an organization to detect, contain, investigate and recover from security incidents. Security policies define expected behavior, responsibilities and controls. Clear reporting channels reduce delays when problems occur.

Learning objectives

- Understand the main purpose of incident response and security policy.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

incident	containment	recovery	policy	reporting
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Incident Response and Security Policy** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Write a simple five-step response to a suspected malware infection.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 41: How the Web Works

Web, Communication and Digital Media

Web browsers request resources from web servers using HTTP or HTTPS. Web pages are commonly built with HTML for structure, CSS for presentation and JavaScript for interactive behavior. Domain names help people reach online services.

Learning objectives

- Understand the main purpose of how the web works.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

browser	server	HTML	CSS	JavaScript
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **How the Web Works** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Describe the roles of HTML, CSS and JavaScript.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 42: Digital Communication

Web, Communication and Digital Media

Email, messaging, video conferencing and collaboration platforms support digital communication. Professional communication uses clear subjects, appropriate tone, careful attachment handling and respect for privacy.

Learning objectives

- Understand the main purpose of digital communication.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

email	messaging	video conference	attachment	netiquette
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Digital Communication** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Write five rules of professional email etiquette.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 43: Social Media and Digital Footprints

Web, Communication and Digital Media

Online activity can create a persistent digital footprint. Users should think carefully before publishing personal information, photos or location details. Organizations also need policies for official accounts and public communication.

Learning objectives

- Understand the main purpose of social media and digital footprints.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

digital footprint	privacy	reputation	sharing	platform
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Social Media and Digital Footprints** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Explain how a digital footprint can affect future opportunities.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 44: Multimedia Formats

Web, Communication and Digital Media

Digital media may include text, images, audio, animation and video. File formats balance quality, compatibility and size. Compression reduces storage and bandwidth requirements, but lossy compression can reduce quality.

Learning objectives

- Understand the main purpose of multimedia formats.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

JPEG	PNG	MP3	MP4	compression
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Multimedia Formats** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Compare lossless and lossy compression with one example each.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 45: Web Accessibility and Usability

Web, Communication and Digital Media

Accessible digital products support people with different abilities and devices. Useful practices include clear headings, keyboard navigation, readable contrast, alternative text and understandable labels. Usability focuses on effectiveness, efficiency and satisfaction.

Learning objectives

- Understand the main purpose of web accessibility and usability.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

accessibility	alt text	contrast	keyboard	usability
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Web Accessibility and Usability** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Identify five accessibility improvements for a school website.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 46: Troubleshooting Method

IT Operations and Support

Effective troubleshooting starts by identifying symptoms, gathering information and forming likely causes. Technicians test one change at a time, document results and confirm that the system works after repair. A structured method reduces unnecessary actions.

Learning objectives

- Understand the main purpose of troubleshooting method.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

symptom	diagnosis	test	solution	documentation
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Troubleshooting Method** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Create a troubleshooting flow for a computer that cannot connect to Wi-Fi.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 47: Preventive Maintenance

IT Operations and Support

Preventive maintenance aims to reduce failures through updates, cleaning, monitoring, backups and replacement planning. Maintenance schedules should consider equipment age, usage and environmental conditions.

Learning objectives

- Understand the main purpose of preventive maintenance.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

maintenance	cleaning	update	monitoring	replacement
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Preventive Maintenance** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Design a monthly maintenance checklist for a computer lab.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 48: User Accounts and Permissions

IT Operations and Support

Accounts identify users and permissions control what they can access or change. The principle of least privilege gives each user only the access required for their role. Shared administrator accounts make accountability difficult and should be avoided.

Learning objectives

- Understand the main purpose of user accounts and permissions.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

account	permission	role	least privilege	administrator
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **User Accounts and Permissions** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Explain why students should not use administrator accounts.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 49: Help Desk and Documentation

IT Operations and Support

A help desk records incidents and service requests, prioritizes work and communicates with users. Good documentation includes problem descriptions, actions taken, solutions and knowledge-base articles for recurring issues.

Learning objectives

- Understand the main purpose of help desk and documentation.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

ticket	priority	SLA	knowledge base	documentation
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Help Desk and Documentation** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Draft a help-desk ticket for a printer that repeatedly jams.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 50: IT Asset Management

IT Operations and Support

Asset management tracks devices, licenses, warranties and assignments throughout their life cycle. Accurate inventories support budgeting, maintenance, security and responsible disposal.

Learning objectives

- Understand the main purpose of it asset management.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

asset	inventory	license	warranty	lifecycle
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **IT Asset Management** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. List the fields that should appear in an IT asset register.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 51: Green IT and E-Waste

IT Operations and Support

Green IT reduces environmental impact through energy efficiency, longer device life, responsible procurement and safe recycling. Electronic waste can contain valuable materials and hazardous substances, so disposal should follow appropriate procedures.

Learning objectives

- Understand the main purpose of green it and e-waste.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

energy efficiency	e-waste	recycling	lifecycle	procurement
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Green IT and E-Waste** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Propose five ways a school can reduce the environmental impact of IT.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 52: Artificial Intelligence in IT

Emerging Technologies and IT in Society

Artificial intelligence systems can classify information, recognize patterns, generate content and support decisions. IT professionals may deploy, monitor and secure AI services. Responsible use requires attention to accuracy, bias, privacy, transparency and human oversight.

Learning objectives

- Understand the main purpose of artificial intelligence in it.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

AI	model	bias	privacy	oversight
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Artificial Intelligence in IT** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Give two benefits and two risks of using AI in education.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 53: Internet of Things

Emerging Technologies and IT in Society

The Internet of Things connects sensors, devices and systems so they can collect data and respond to conditions. Examples include smart meters, industrial sensors and environmental monitoring. Security is important because connected devices can create new attack surfaces.

Learning objectives

- Understand the main purpose of internet of things.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

IoT	sensor	device	telemetry	security
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Internet of Things** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Design an IoT idea for monitoring a school environment.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 54: Virtualization and Containers

Emerging Technologies and IT in Society

Virtualization allows multiple virtual machines to share physical hardware, while containers package applications with required dependencies in a lightweight way. These technologies improve utilization, isolation and deployment flexibility.

Learning objectives

- Understand the main purpose of virtualization and containers.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

virtual machine	hypervisor	container	isolation	deployment
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Virtualization and Containers** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Compare a virtual machine and a container.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 55: Digital Transformation

Emerging Technologies and IT in Society

Digital transformation changes processes, services and organizational culture through technology. Successful transformation is not only about buying software; it also requires leadership, training, process redesign and measurable goals.

Learning objectives

- Understand the main purpose of digital transformation.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

process	service	change	training	measurement
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Digital Transformation** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Give an example of digital transformation in a school or public service.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Lesson 56: Ethics and Responsible Technology

Emerging Technologies and IT in Society

Technology decisions can affect privacy, access, employment, fairness and the environment. Ethical IT practice considers stakeholders, potential harms, transparency and accountability. Professionals should follow laws, policies and recognized codes of conduct.

Learning objectives

- Understand the main purpose of ethics and responsible technology.
- Use correct technical vocabulary in explanations.
- Apply the concept to a practical school, home or workplace scenario.

Key terms

ethics	fairness	privacy	accountability	stakeholder
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Explanation and application

This topic is important because IT decisions affect performance, cost, security and user experience. A good solution begins with clear requirements, then compares alternatives and checks whether the chosen technology can be supported over time.

In practical work, students should avoid memorizing isolated definitions. Instead, connect each idea to a real system: who uses it, what data it handles, what could fail, and what controls or procedures keep the system reliable.

Worked example

A school is reviewing its technology. The IT team uses the ideas from **Ethics and Responsible Technology** to compare needs, identify constraints and choose a solution. The team records assumptions, tests the result and documents the final configuration so that future support is easier.

Practice

1. Discuss one ethical issue created by automated decision systems.
2. Define the five key terms above in one sentence each.
3. Give one advantage, one limitation and one real-life use connected to this lesson.

Study note

For examination answers, use the pattern: **identify -> explain -> apply -> justify**. A strong technical answer does more than name a feature; it explains why that feature matters in the given situation.

Practical Lab: Build a PC Specification

Create three computer specifications: office work, graphic design and school laboratory. For each, justify processor, RAM, storage, monitor and network requirements.

Task steps

1. Read the scenario and define user requirements.
2. Choose suitable tools, settings or components.
3. Create the required output or design.
4. Test your work and record evidence.
5. Write a short reflection explaining what could be improved.

Assessment criteria

Criterion	Excellent evidence
Technical correctness	Choices and procedures are accurate.
Justification	Decisions are clearly linked to requirements.
Security and safety	Relevant risks and controls are considered.
Presentation	Work is organized and easy to follow.

Reflection questions

What worked well? What problem did you solve? Which assumption was most important? What security or usability issue should be checked before real deployment?

Practical Lab: Spreadsheet Budget

Build a monthly budget with income, expenses, totals, percentage calculations and a chart. Use formulas rather than typing calculated results.

Task steps

1. Read the scenario and define user requirements.
2. Choose suitable tools, settings or components.
3. Create the required output or design.
4. Test your work and record evidence.
5. Write a short reflection explaining what could be improved.

Assessment criteria

Criterion	Excellent evidence
Technical correctness	Choices and procedures are accurate.
Justification	Decisions are clearly linked to requirements.
Security and safety	Relevant risks and controls are considered.
Presentation	Work is organized and easy to follow.

Reflection questions

What worked well? What problem did you solve? Which assumption was most important? What security or usability issue should be checked before real deployment?

Practical Lab: Database Mini Project

Design a small library database with books, borrowers and loans. Identify primary and foreign keys and write sample queries.

Task steps

1. Read the scenario and define user requirements.
2. Choose suitable tools, settings or components.
3. Create the required output or design.
4. Test your work and record evidence.
5. Write a short reflection explaining what could be improved.

Assessment criteria

Criterion	Excellent evidence
Technical correctness	Choices and procedures are accurate.
Justification	Decisions are clearly linked to requirements.
Security and safety	Relevant risks and controls are considered.
Presentation	Work is organized and easy to follow.

Reflection questions

What worked well? What problem did you solve? Which assumption was most important? What security or usability issue should be checked before real deployment?

Practical Lab: Small Network Plan

Draw a network for a small office with ten computers, a printer, Wi-Fi, a router and Internet access. Label devices and explain traffic flow.

Task steps

1. Read the scenario and define user requirements.
2. Choose suitable tools, settings or components.
3. Create the required output or design.
4. Test your work and record evidence.
5. Write a short reflection explaining what could be improved.

Assessment criteria

Criterion	Excellent evidence
Technical correctness	Choices and procedures are accurate.
Justification	Decisions are clearly linked to requirements.
Security and safety	Relevant risks and controls are considered.
Presentation	Work is organized and easy to follow.

Reflection questions

What worked well? What problem did you solve? Which assumption was most important? What security or usability issue should be checked before real deployment?

Practical Lab: Cybersecurity Audit

Inspect a hypothetical school system. Identify weak passwords, missing backups, unpatched software and unsafe sharing. Propose controls and priorities.

Task steps

1. Read the scenario and define user requirements.
2. Choose suitable tools, settings or components.
3. Create the required output or design.
4. Test your work and record evidence.
5. Write a short reflection explaining what could be improved.

Assessment criteria

Criterion	Excellent evidence
Technical correctness	Choices and procedures are accurate.
Justification	Decisions are clearly linked to requirements.
Security and safety	Relevant risks and controls are considered.
Presentation	Work is organized and easy to follow.

Reflection questions

What worked well? What problem did you solve? Which assumption was most important? What security or usability issue should be checked before real deployment?

Practical Lab: Website Planning

Plan a four-page school website. Define audience, navigation, content, accessibility requirements and a consistent visual structure.

Task steps

1. Read the scenario and define user requirements.
2. Choose suitable tools, settings or components.
3. Create the required output or design.
4. Test your work and record evidence.
5. Write a short reflection explaining what could be improved.

Assessment criteria

Criterion	Excellent evidence
Technical correctness	Choices and procedures are accurate.
Justification	Decisions are clearly linked to requirements.
Security and safety	Relevant risks and controls are considered.
Presentation	Work is organized and easy to follow.

Reflection questions

What worked well? What problem did you solve? Which assumption was most important? What security or usability issue should be checked before real deployment?

Practical Lab: Help Desk Simulation

Write three help-desk tickets: no Internet, slow computer and printer problem. Include symptoms, questions, tests and final resolution.

Task steps

1. Read the scenario and define user requirements.
2. Choose suitable tools, settings or components.
3. Create the required output or design.
4. Test your work and record evidence.
5. Write a short reflection explaining what could be improved.

Assessment criteria

Criterion	Excellent evidence
Technical correctness	Choices and procedures are accurate.
Justification	Decisions are clearly linked to requirements.
Security and safety	Relevant risks and controls are considered.
Presentation	Work is organized and easy to follow.

Reflection questions

What worked well? What problem did you solve? Which assumption was most important? What security or usability issue should be checked before real deployment?

Practical Lab: Cloud Collaboration

Create a workflow for a team preparing a report using shared storage, version control, permissions and online meetings.

Task steps

1. Read the scenario and define user requirements.
2. Choose suitable tools, settings or components.
3. Create the required output or design.
4. Test your work and record evidence.
5. Write a short reflection explaining what could be improved.

Assessment criteria

Criterion	Excellent evidence
Technical correctness	Choices and procedures are accurate.
Justification	Decisions are clearly linked to requirements.
Security and safety	Relevant risks and controls are considered.
Presentation	Work is organized and easy to follow.

Reflection questions

What worked well? What problem did you solve? Which assumption was most important? What security or usability issue should be checked before real deployment?

Case Study: IT in Banking

Study how information systems support accounts, payments, ATMs, mobile banking, fraud monitoring and customer service. Identify security and availability requirements.

Discussion framework

Identify the users, data, hardware, software, network services and security controls. Then explain the consequences of downtime, inaccurate data or unauthorized access.

Question	Focus
1	What information is collected and why?
2	Which systems or services process it?
3	What are the main risks?
4	Which controls improve reliability and security?
5	How should success be measured?

Case Study: IT in Health Care

Explore electronic records, appointment systems, medical imaging and telemedicine. Discuss privacy, reliability and access.

Discussion framework

Identify the users, data, hardware, software, network services and security controls. Then explain the consequences of downtime, inaccurate data or unauthorized access.

Question	Focus
1	What information is collected and why?
2	Which systems or services process it?
3	What are the main risks?
4	Which controls improve reliability and security?
5	How should success be measured?

Case Study: IT in Education

Examine learning platforms, digital libraries, school management systems and remote learning. Evaluate benefits, barriers and teacher support needs.

Discussion framework

Identify the users, data, hardware, software, network services and security controls. Then explain the consequences of downtime, inaccurate data or unauthorized access.

Question	Focus
1	What information is collected and why?
2	Which systems or services process it?
3	What are the main risks?
4	Which controls improve reliability and security?
5	How should success be measured?

Case Study: IT in Government

Consider digital identity, online forms, tax systems and public information portals. Discuss accessibility, security and inclusion.

Discussion framework

Identify the users, data, hardware, software, network services and security controls. Then explain the consequences of downtime, inaccurate data or unauthorized access.

Question	Focus
1	What information is collected and why?
2	Which systems or services process it?
3	What are the main risks?
4	Which controls improve reliability and security?
5	How should success be measured?

Case Study: IT in Business

Map an information flow from sales to inventory, accounting and management reporting. Explain how integrated systems reduce duplication.

Discussion framework

Identify the users, data, hardware, software, network services and security controls. Then explain the consequences of downtime, inaccurate data or unauthorized access.

Question	Focus
1	What information is collected and why?
2	Which systems or services process it?
3	What are the main risks?
4	Which controls improve reliability and security?
5	How should success be measured?

Revision 1: Foundations and Hardware

Review IT concepts, components, CPU, memory, storage, input/output, interfaces and safe use.

Quick-check questions

1. Define five key terms from this unit.
2. Explain one advantage and one limitation of a technology from this unit.
3. Describe a realistic problem and the IT solution you would recommend.
4. Identify one security or privacy issue.
5. Write one comparison using technical criteria such as cost, speed, reliability or usability.

Self-assessment

Skill	Not yet	Developing	Confident
Definitions			
Explanations			
Practical application			
Justification			
Security awareness			

Revision 2: Software and Productivity

Review operating systems, file management, licensing, word processing, spreadsheets, presentations and collaboration.

Quick-check questions

1. Define five key terms from this unit.
2. Explain one advantage and one limitation of a technology from this unit.
3. Describe a realistic problem and the IT solution you would recommend.
4. Identify one security or privacy issue.
5. Write one comparison using technical criteria such as cost, speed, reliability or usability.

Self-assessment

Skill	Not yet	Developing	Confident
Definitions			
Explanations			
Practical application			
Justification			
Security awareness			

Revision 3: Databases and Networks

Review relational databases, validation, SQL, LAN/WAN, network devices, protocols, wireless networking and cloud services.

Quick-check questions

1. Define five key terms from this unit.
2. Explain one advantage and one limitation of a technology from this unit.
3. Describe a realistic problem and the IT solution you would recommend.
4. Identify one security or privacy issue.
5. Write one comparison using technical criteria such as cost, speed, reliability or usability.

Self-assessment

Skill	Not yet	Developing	Confident
Definitions			
Explanations			
Practical application			
Justification			
Security awareness			

Revision 4: Cybersecurity

Review the CIA model, authentication, malware, phishing, backups, privacy, security policies and incident response.

Quick-check questions

1. Define five key terms from this unit.
2. Explain one advantage and one limitation of a technology from this unit.
3. Describe a realistic problem and the IT solution you would recommend.
4. Identify one security or privacy issue.
5. Write one comparison using technical criteria such as cost, speed, reliability or usability.

Self-assessment

Skill	Not yet	Developing	Confident
Definitions			
Explanations			
Practical application			
Justification			
Security awareness			

Revision 5: Web, Support and Emerging Tech

Review web technology, digital communication, accessibility, troubleshooting, asset management, AI, IoT and ethics.

Quick-check questions

1. Define five key terms from this unit.
2. Explain one advantage and one limitation of a technology from this unit.
3. Describe a realistic problem and the IT solution you would recommend.
4. Identify one security or privacy issue.
5. Write one comparison using technical criteria such as cost, speed, reliability or usability.

Self-assessment

Skill	Not yet	Developing	Confident
Definitions			
Explanations			
Practical application			
Justification			
Security awareness			

Mock Examination A

Answer structured questions covering hardware, software, databases, networks and cybersecurity. Show reasoning and use correct technical vocabulary.

Section A - Short answers

1. Define a database primary key. **2.** State two functions of an operating system. **3.** Explain one difference between a router and a switch. **4.** Give two reasons for keeping backups. **5.** State one benefit and one risk of cloud software.

Section B - Scenario

A growing organization needs new computers, shared files, reliable Internet access and improved security. Recommend a suitable solution. Justify hardware, software, networking, backup and user-account decisions.

Section C - Extended response

Discuss how information technology can improve services while also creating risks related to privacy, cybersecurity, inequality or environmental impact. Use examples.

Mock Examination B

Complete scenario-based questions involving a school, small business and public service organization. Recommend appropriate IT solutions and justify choices.

Section A - Short answers

1. Define a database primary key. **2.** State two functions of an operating system. **3.** Explain one difference between a router and a switch. **4.** Give two reasons for keeping backups. **5.** State one benefit and one risk of cloud software.

Section B - Scenario

A growing organization needs new computers, shared files, reliable Internet access and improved security. Recommend a suitable solution. Justify hardware, software, networking, backup and user-account decisions.

Section C - Extended response

Discuss how information technology can improve services while also creating risks related to privacy, cybersecurity, inequality or environmental impact. Use examples.

Answer Guide: Core Concepts

Use this page to check key definitions and short-answer expectations. Strong answers identify the concept, explain how it works and connect it to the scenario.

Model-answer principles

Short answers should use precise technical vocabulary. Scenario answers should connect each recommendation to a requirement. Extended answers should consider more than one side, use examples and reach a reasoned conclusion.

Example

Question: Why use multi-factor authentication?

Strong answer: MFA requires evidence from more than one factor, so a stolen password alone is less likely to give an attacker access to the account.